Servicenow

1. Flow Designer Architecture

Flow Designer is ServiceNow's **no-code / low-code automation engine** used to automate business processes.

Core Components:

1. Trigger (Start Point)

Defines **when the flow starts**

Types:

•Record Trigger (on table changes)

•Scheduled Trigger (time-based)

•Application Trigger (e.g., Service Catalog)

•Event Trigger (event fired)

1. Flow Designer Architecture

Core Components:

2. Actions (What Happens)

Steps executed after the trigger:

Create / Update Record

Send Email / Notification

Ask for Approval

Call REST API

Script Logic

1. Flow Designer Architecture

Core Components:

3. Data Flow

How data moves inside the flow:

Data Pills (drag & drop values)

Inputs / Outputs

Flow Variables

2. Record Triggers (VERY IMPORTANT)

This is the most used trigger in real projects.

“When to Run Flow” Options (Critical)

Run Trigger Options

✓ Only if not currently running

Prevents infinite loops (recursion)

✓ For each unique change

Runs for every update

2. Record Triggers (VERY IMPORTANT)

This is the most used trigger in real projects.

When to Run (Execution Timing)

- **Before**

❖ Runs **before record is saved**

❖ Faster

Use cases:

- Validation
- Auto-fill fields

2. Record Triggers (VERY IMPORTANT)

When to Run (Execution Timing)

- **After (Most Common)**

❖ Runs **after record is saved**

Use cases:

- Notifications
- Creating related records
- Integrations

2. Record Triggers (VERY IMPORTANT)

When to Run (Execution Timing)

- **Async (Asynchronous)**

❖ Runs in background

❖ Does NOT affect user performance

Use cases:

- External APIs
- Heavy processing
- Large data operations

2. Record Triggers (VERY IMPORTANT)

When to Run (Execution Timing)

- **Conditions**

❖ Defines **when flow should execute**

Examples:

- State changes to “Closed”
- Priority is High

Advanced Options

✓ **Run only if conditions are met**

•Prevent unnecessary execution

✓ **Run after update only if fields change**

•Optimize performance

✓ **Repeat execution**

•Decide if flow should run multiple times

3. Subflows (Reusable Logic)

What is a Subflow?
A **reusable mini-flow**

When to Use
- Repeated logic
- Approval processes
- Notifications
- Shared integrations

Benefits
✓ Reusability
✓ Clean architecture
✓ Easy maintenance

3. Subflows (Reusable Logic)

- **Structure**

Inputs:

- Record
- User
- Variables

Outputs:

- Status
- Result
- Data

4. Custom Flow Actions

A **custom reusable action** you build

Structure

1. Inputs
2. Logic (script / steps)
3. Outputs

Example

Action: **Calculate Discount**

Inputs:

- Price
- Discount %

Output:

- Final Price

4. Custom Flow Actions

When to Use

✓ Complex logic
✓ Integration with external systems
✓ Reusable across multiple flows

5. Service Catalog Integration (Enterprise Core)

This is where real business automation happens

Data Flow Structure

```
Request (REQ)
    ↓
Requested Item (RITM)
    ↓
Catalog Task (TASK)
```

5. Service Catalog Integration (Enterprise Core)

How Flow Works Here

1. User submits a Catalog Item
2. System creates:
   - Request
   - RITM
3. Flow is triggered
4. Tasks are created
5. Approvals handled
6. Fulfillment completed

5. Service Catalog Integration (Enterprise Core)

Trigger Types

- Service Catalog Trigger

  OR

- Record Trigger on:

  sc_req_item

Real Use Cases

•Laptop Request
•Employee Onboarding
•Access Requests

6. Error Handling & Debugging

Critical in Enterprise Systems

Error Handling

✓ **Run if Failed**

•Handle failure of a step

✓ **Error Handler Step**

•Define fallback logic

6. Error Handling & Debugging

- **Rollback**

Undo operations if something fails

- **Logging**

System Logs
Flow Execution Logs

- **Debugging Tools**

✓ **Execution Details**
Shows:
Each step
Inputs / Outputs
Errors

- **Test Mode**

Run flow manually for testing

Practice

1. Smart Laptop Recommendation (AI-like Logic)

Scenario:

An employee submits a laptop request, but the system automatically selects the most suitable device based on their role.

Requirement:

- Developer → High-performance laptop
- HR → Basic laptop
- Designer → GPU laptop

2. Budget Control with Hard Stop

Scenario:

The company has a limited monthly budget for laptop purchases.

Requirement:

If budget is exceeded → Reject request

If budget is near limit → Require Finance approval

3. Auto Asset Allocation

Scenario:

Instead of manual selection, the system automatically assigns an available laptop from inventory.

Requirement:

If available → Assign asset

If not → Create procurement task

4. Multi-Level Approval Based on Risk

Scenario:

Approval is not only based on cost but also device sensitivity.

Requirement:

Normal device → Manager approval

High-end device → Manager + IT Director

Special device → Security approval

5. Auto Escalation Before SLA Breach

Scenario:

IT must deliver laptops within SLA.

Requirement:

Before SLA breach → Send reminder

After SLA breach → Escalate

6. Employee Exit Automation

Scenario:

An employee leaves the company.

Requirement:

Collect laptop

Disable access

Update asset status

7. Duplicate Request Prevention

Scenario:

An employee submits multiple laptop requests.

Requirement:

Prevent request if another active request exists

8. Reusable Notification Engine (Subflow)

Scenario:

Multiple flows require different types of notifications.

Requirement:

Create a reusable Subflow:

Inputs:

- User
- Notification type (approval / rejection / ready)

9. Conditional Task Splitting

Scenario:

A request includes multiple devices.

Requirement:

Each device → separate task

10. Performance-Based Assignment

Scenario:

Tasks should be assigned intelligently, not randomly.

Requirement:

Assign task based on:

Lowest workload

Best performance

Student Practice

Scenario:

Manager submits:

👉 “New Branch Setup Request”

Flow Steps:

1. Trigger from Service Catalog
2. Approval from Admin
3. Create Tasks:
    1. IT Setup
    2. Kitchen Setup
    3. Staff Hiring
4. Send Notifications
5. Integrate with external system

What We Used

Component	Usage
Trigger	Catalog
Flow	Main process
Subflow	Approval logic
Action	Integration
Error Handling	Logging + retry

1. Smart Chef Assignment (Using GlideRecord)

Scenario:

When a new food order is created, assign it to the **best available chef** based on workload.

Requirement:

•Assign to chef with:

- Lowest active orders
- Matching specialization (e.g., grill, desserts)

1. Smart Chef Assignment (Using GlideRecord)

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•Assign to chef with:

- Lowest active orders
- Matching specialization (e.g., grill, desserts)

2. Auto-Cancel Unpaid Orders (Using GlideDateTime)

Scenario:

If a customer doesn’t pay within 15 minutes → order is canceled automatically

Requirement:

After 15 minutes:

If payment status = Pending → Cancel order

3. Dynamic Discount Engine (GlideRecord + Logic)

Scenario:

Apply discounts based on customer history

Requirement:

If customer ordered > 10 times → 10% discount

20 times → 20% discount

4. Ingredient Stock Check (GlideRecord Query)

Scenario:

Before confirming an order, check if ingredients are available

Requirement:

If any ingredient is out of stock → block order

5. Auto-Escalation for Delayed Orders (GlideDateTime)

Scenario:

If order preparation takes too long → notify manager

Requirement:

After 20 minutes:

If not completed → escalate

6. Prevent Duplicate Orders (GlideRecord)

Scenario:

Customer submits same order twice quickly

Requirement:

Prevent duplicate orders within 5 minutes

Scenario 1: Order Creation

- When a new order is created:
- - Auto-fill order type (Before)
- - Notify kitchen (After)

Solution

- Trigger: Record (Order)
- Before: Set order category
- After: Send notification to kitchen

Scenario 2: Prevent Duplicate Orders

- Customer submits same order twice quickly

Solution

- Trigger: Before Insert
- Condition: Same customer + same dish
- Action: Stop execution

Scenario 3: Avoid Infinite Loop

- Flow updates order → triggers itself again

Solution

- Use: 'Only if not currently running'
- Prevents recursion

Scenario 4: Run Only on Status Change

- Run flow only when order becomes 'Ready'

Solution

- Condition: State changes to Ready
- Avoid running on other updates

Scenario 5: Async for Heavy Tasks

- After order ready:
- - Send SMS
- - Call external delivery API

Solution

- Use Async trigger
- Runs in background without slowing system

Scenario 6: Scheduled Reminder

- Every day:
- - Check pending orders
- - Notify staff

Solution

- Trigger: Scheduled (daily)
- Action: Send reminder

Scenario 7: Conditional Execution

- VIP orders need special handling

Solution

- Trigger Condition: Order type = VIP
- Inside Flow: Create extra tasks

Scenario 8: Field Change Optimization

- Run flow only when payment status changes

Solution

- Advanced Option:
- Run only if Payment field changes

Outcome

- ✓ Efficient triggers
- ✓ Better performance
- ✓ Controlled automation
- ✓ Real restaurant workflows

Thank you

Questions

Q1: Which of the following are valid components or requirements for an automated Certificate Renewal flow in ServiceNow? (Select all that Apply)

A. Routing Policy

B. Certificate Signing Request (CSR)

C. PKI Admin role

D. Flow Designer

E. MID Server with access to the Certificate Authority (CA)

Answer A, B, C, D, E

- ✅ **Routing Policy:** Essential for determining which CA handles the request.
- ✅ **CSR:** Mandatory for the renewal process to be initiated.
- ✅ **PKI Admin role:** Required to manage and trigger these specific certificate flows.
- ✅ **Flow Designer:** The engine where automated certificate flows reside.
- ✅ **MID Server:** Necessary for communication between your instance and internal/external CAs

Q2: What occurs when a certificate discovery workflow identifies a certificate that is missing an "Owned By" or "Assigned To" value? (Select all that Apply)

A. The system automatically deletes the record to maintain data integrity.

B. A message is logged recommending a Discovery run to populate details.

C. The system may prevent automatic renewal task creation if specific group fields are empty.

D. The certificate is automatically assigned to the 'admin' user

Answer B,C

- ✅ B: If details like CA or Order ID are missing, the system logs a message suggesting Discovery.
- ✅ C: Automated tasks rely on group/owner fields; if empty, the workflow may not know where to route the task.
- ❌ A & D: ServiceNow does not delete CIs automatically for missing values, nor does it default assignments to 'admin' without custom logic.

Q3: In Flow Designer, which of these elements can be used to pass data between a main flow and a subflow used for certificate revocation? (Select all that Apply)

A. Inputs

B. Outputs

C. Data Pills

D. Update Sets

Answer A,B,C

- ✅ Inputs/Outputs: These are the primary methods for moving data between flows and subflows.
- ✅ Data Pills: These represent the actual runtime variables passed between steps.
- ❌ Update Sets: Used to move configuration between environments, not data between flow steps

Q4: Which ServiceNow application is primarily used to build and manage modern certificate lifecycle workflows?

A. Workflow Editor

B. Flow Designer

C. Service Portal

D. VTB (Visual Task Board)

Answer B

•ServiceNow has shifted its automation focus from the legacy Workflow Editor to Flow Designer for modern applications like Certificate Management.

Q5: Which of the following are benefits of using the automated "Renew Certificate" flow over manual processes? (Select all that Apply)

A. Reduces the risk of human error during CSR generation.

B. Automatically updates the Order ID in the Certificate Task table.

C. Guarantees that every certificate in the network is discovered instantly.

D. Ensures change management governance by following pre-defined routing policies.

Answer A,B,D

- ✅ A: Automation removes manual steps where typos often occur.
- ✅ B: The flow automatically writes back the Order ID to the relevant [sn_disco_certmgmt_certificate_task] tables.
- ✅ D: Automated flows follow logic that forces compliance with internal policies.
- ❌ C: Discovery is a scheduled or manual process; flows manage the renewal, but discovery find the certs first

Thank you